

SPECIAL RELATIVITY

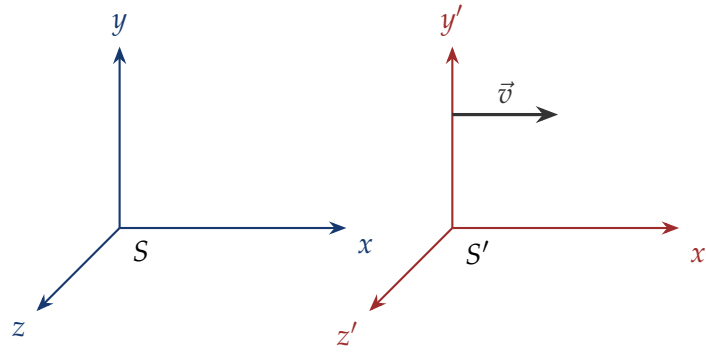
Lorentz Boosts in Matrix Form

From the standard boost to a boost in any direction

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1 The standard boost along x

Frame S' moves with velocity v along the common x -axis of S and S' .



The transformation equations are

$$\begin{aligned} x' &= \frac{x - vt}{\sqrt{1 - v^2/c^2}}, & y' &= y, & z' &= z, \\ t' &= \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}, \end{aligned} \quad (1)$$

with

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \quad \beta = \frac{v}{c}. \quad (2)$$

1.1 Rewriting with $x^0 = ict$

Using $ct = x^0/i = -ix^0$, the spatial part becomes

$$\begin{aligned} x' &= \gamma(x - vt) \\ &= \gamma\left(x - \frac{v}{c}ct\right) \\ &= \gamma(x - \beta ct) \\ &= \gamma\left(x - \beta \frac{ct i}{i}\right) \\ &= \gamma(x + i\beta x^0). \end{aligned} \quad (3)$$

For the time part, $t' = \gamma(t - vx/c^2)$, so

$$\begin{aligned} x'_0 &= ict' = ic \gamma \left(t - \frac{vx}{c^2} \right) \\ &= \gamma \left(ict - i \frac{vx}{c} \right) \\ &= \gamma (x_0 - i\beta x). \end{aligned} \quad (4)$$

2 Matrix form

Collecting the results, with $x_0 = ict$, $x_1 = x$, $x_2 = y$, $x_3 = z$:

$$\begin{pmatrix} x'_0 \\ x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} \gamma & -i\beta\gamma & 0 & 0 \\ i\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}. \quad (5)$$

$$x'_\mu = \Lambda_{\mu\nu} x_\nu, \quad \mu, \nu = 0, 1, 2, 3.$$

3 Boost in an arbitrary direction

Decompose the position vector relative to the direction of motion:

$$\vec{x} = \vec{x}_\perp + \vec{x}_\parallel, \quad (6)$$

where

- $\vec{x}_\perp$ is the component of $\vec{x}$ *perpendicular* to the direction of motion,
- $\vec{x}_\parallel$ is the component of $\vec{x}$ *parallel* to the direction of motion.

Only the parallel part is affected by the boost:

$$\begin{aligned} \vec{x}' &= \vec{x}_\perp + \frac{\vec{x}_\parallel - \vec{v}t}{\sqrt{1 - v^2/c^2}} \\ &= (\vec{x} - \vec{x}_\parallel) + \gamma(\vec{x}_\parallel - \vec{v}t) \\ &= \vec{x} + (\gamma - 1)\vec{x}_\parallel - \gamma\vec{v}t. \end{aligned} \quad (7)$$

The parallel component is the projection of $\vec{x}$ onto $\hat{e} = \vec{v}/|\vec{v}|$:

$$\vec{x}_\parallel = |\vec{x}_\parallel| \hat{e} = \frac{\vec{x} \cdot \vec{v}}{|\vec{v}|} \frac{\vec{v}}{|\vec{v}|} = \frac{(\vec{x} \cdot \vec{v}) \vec{v}}{|\vec{v}|^2}. \quad (8)$$

Hence

$$\vec{x}' = \vec{x} + (\gamma - 1) \frac{(\vec{x} \cdot \vec{v}) \vec{v}}{|\vec{v}|^2} - \gamma \vec{v}t. \quad (9)$$

3.1 Introducing $\vec{\beta} = \vec{v}/c$

Since $(\vec{x} \cdot \vec{v}) \vec{v}/|\vec{v}|^2 = (\vec{x} \cdot \vec{\beta}) \vec{\beta}/|\vec{\beta}|^2$, and $\gamma \vec{v}t = \gamma \vec{\beta} ct = \gamma \vec{\beta} \frac{x^0}{i} = -i\gamma \vec{\beta} x^0$,

$$\boxed{\vec{x}' = \vec{x} + (\gamma - 1) \frac{(\vec{x} \cdot \vec{\beta}) \vec{\beta}}{|\vec{\beta}|^2} + i\gamma \vec{\beta} x^0} \quad (10)$$

with $\vec{x}' = x'_1\hat{i} + x'_2\hat{j} + x'_3\hat{k}$, $\vec{x} = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$ and $\vec{\beta} = \beta_1\hat{i} + \beta_2\hat{j} + \beta_3\hat{k}$.

3.2 The time component

$$t' = \frac{t - \vec{v} \cdot \vec{x}/c^2}{\sqrt{1 - v^2/c^2}} = \gamma \left(t - \frac{\vec{\beta} \cdot \vec{x}}{c} \right) = \frac{\gamma}{c} (ct - \vec{\beta} \cdot \vec{x}), \quad (11)$$

and multiplying by ic ,

$$\begin{aligned} x'_0 &= ict' = i\gamma (ct - \vec{\beta} \cdot \vec{x}) \\ &= \gamma (x^0 - i\vec{\beta} \cdot \vec{x}) \\ &= \gamma x_0 - i\gamma\beta_1 x_1 - i\gamma\beta_2 x_2 - i\gamma\beta_3 x_3. \end{aligned} \quad (12)$$

4 The general boost matrix

In components, the spatial part of the boost reads

$$x'_i = x_i + (\gamma - 1) \frac{\beta_i (\vec{\beta} \cdot \vec{x})}{|\vec{\beta}|^2} + i\gamma \beta_i x_0, \quad i = 1, 2, 3, \quad (13)$$

and together with the time component this gives the full matrix:

$$\begin{pmatrix} x'_0 \\ x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} \gamma & -i\gamma\beta_1 & -i\gamma\beta_2 & -i\gamma\beta_3 \\ i\gamma\beta_1 & 1 + \frac{\gamma-1}{|\vec{\beta}|^2}\beta_1^2 & \frac{\gamma-1}{|\vec{\beta}|^2}\beta_1\beta_2 & \frac{\gamma-1}{|\vec{\beta}|^2}\beta_1\beta_3 \\ i\gamma\beta_2 & \frac{\gamma-1}{|\vec{\beta}|^2}\beta_1\beta_2 & 1 + \frac{\gamma-1}{|\vec{\beta}|^2}\beta_2^2 & \frac{\gamma-1}{|\vec{\beta}|^2}\beta_2\beta_3 \\ i\gamma\beta_3 & \frac{\gamma-1}{|\vec{\beta}|^2}\beta_1\beta_3 & \frac{\gamma-1}{|\vec{\beta}|^2}\beta_2\beta_3 & 1 + \frac{\gamma-1}{|\vec{\beta}|^2}\beta_3^2 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}. \quad (14)$$

Special case. When the velocity is along the x -axis only, i.e. $\beta_1 = \beta$ and $\beta_2 = \beta_3 = 0$, this reduces to the ordinary Lorentz transformation, Eq. (5), since $1 + \frac{\gamma-1}{\beta^2}\beta^2 = \gamma$.

5 Lorentz scalar and the Minkowski metric

Under a Lorentz transformation, a vector with lower and upper indices transforms as

$$x_\mu \longrightarrow x'_\mu = \Lambda_{\mu\nu} x^\nu, \quad (15)$$

$$x^\mu \longrightarrow x'^\mu = \Lambda^{\mu\nu} x_\nu = \Lambda^\mu{}_\nu x^\nu. \quad (16)$$

Convention. As in the previous notes, in this section $x^0 = ct$ is the real time coordinate, and the minus signs of the interval are carried by the index-lowering prescription (equivalently, by the metric).

The square of x is a **Lorentz scalar**:

★ Important: the Lorentz scalar

$$\begin{aligned} x^2 &= x_\mu x^\mu = x^0 x_0 + x^1 x_1 + x^2 x_2 + x^3 x_3 \\ &= x^0 x^0 - x^1 x^1 - x^2 x^2 - x^3 x^3 \quad (x_0 = x^0, x_i = -x^i) \\ &= (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2. \end{aligned}$$

For the opposite signature, $(-, +, +, +)$, the result is $x^2 = -(x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2$.

5.1 Inserting a metric

The index-lowering signs can be packaged into a matrix, the **Minkowski metric** $\eta_{\mu\nu}$, so that $x^2 = \eta_{\mu\nu} x^\mu x^\nu$:

$$\eta_{\mu\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (\text{Minkowski matrix}) \quad (17)$$

or, in the opposite convention,

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix} \quad (18)$$

In either case the metric is a 4×4 matrix.